Project Design Phase
Solution Architecture

Date	15/10/2026
Team ID	
Project Name	Metro Ticket Booking System using ServiceNow
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process - with many sub-processes - that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

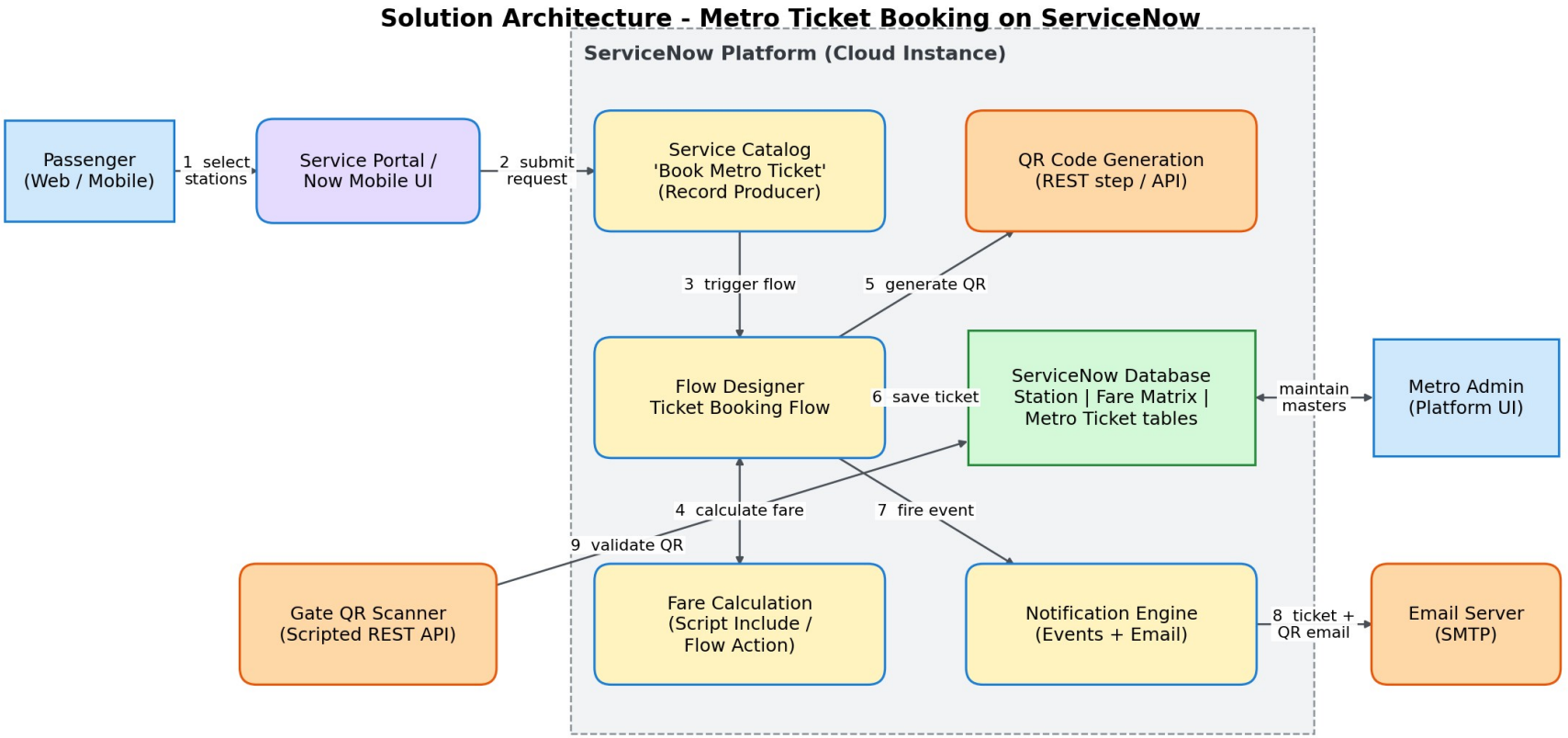


Figure 1: Architecture and data flow of the Metro Ticket Booking System on ServiceNow

Data flow (numbered as in the diagram):

Step	Description	Component
1	The passenger selects source and destination stations, date and passengers in the portal or mobile app.	Service Portal / Now Mobile
2	The form is submitted as a request for the 'Book Metro Ticket' catalog item.	Service Catalog

Step	Description	Component
3	The submitted request triggers the ticket booking flow.	Flow Designer
4	The flow looks up the fare for the selected stations and calculates the total fare.	Fare calculation (Script Include / Flow Action)
5	A unique QR code is generated for the ticket.	QR generation (REST step / API)
6	The ticket record with fare, status and QR code is saved.	ServiceNow database
7	An event is fired when the ticket is created.	Notification engine
8	A confirmation email with ticket details and QR code is sent to the passenger.	Email server (SMTP)
9	At the gate, the QR code is scanned and the ticket is validated and marked as used.	Scripted REST API
-	The administrator maintains stations and fares in the platform.	Platform UI

Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/ (sample format)